Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18RADE742]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

INDUSTRY 4.0

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	List and explain the various building blocks or components of industry 4.0.	**06**
	b.	Briefly discuss on various phases of industrial revolution with an example in each phase.	**08**
2.	**a.**	Explain the significance of Industrial Internet of Things (IIoT) in industry 4.0 and mention the main requirements for communication and networking.	**07**
	b.	Explain the concepts of Adaptive robotics and mention the general characteristics of Adaptive robotics applications.	**07**
3.	**a.**	Define roadmap in industry 4.0 frame work. Explain the stages of proposed framework in the new product and process development phase with a neat design flow.	**08**
	b.	List all the key benefits of technology road map in industry 4.0.	**06**
4.	**a.**	Explain the strategy phase of technology road map in industry 4.0.	**07**
	b.	Compare between lean and traditional manufacturing.	**07**
5.	**a.**	Explain the significance of sensor and camera technologies in obtaining decisive information about the industrial tasks.	**08**
	b.	Explain the capabilities of cloud robotics in smart factories.	**06**
6.	**a.**	Define Internet of Robotic Things. List and explain the various layers involved in its architecture.	**08**
	b.	Elaborate on the components of collaborative robots with major integrated entities.	**06**
7.	**a.**	Explain the impact of inefficient digital strategy with lack of resource in an industry 4.0 framework.	**07**
	b.	Explain how Industry 4.0 technologies are affected due to substantial investment under financing conditions.	**07**
8.	**a.**	List and explain the various legal challenges under Industry 4.0 legal framework	**08**
	b.	Discuss the role of data security and standards for small and medium enterprises to switch to new Industry 4.0 technologies.	**06**